

Vineeth Sakhamuru

813-380-3294 | sakhamuruvineeth@gmail.com | [LinkedIn](#) | [GitHub](#) | U.S. Permanent Resident

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

ML/AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, Transformers, BERT, GPT-3.5/4, LLMs, RAG, Fine-tuning, Prompt Engineering, NLP, Deep Learning, sentence-transformers

Cloud & Tools: AWS (SageMaker, Lambda, Textract, EC2), Azure (ML, Document Intelligence), GCP, Docker, Kubernetes, Git, GitLab CI/CD, MLOps, pytest

Databases: PostgreSQL, MongoDB, SQLite, FAISS, Pinecone, ChromaDB, Weaviate

Web: FastAPI, Flask, Streamlit, REST API, Microservices

EDUCATION

Northeastern University

Master of Science in Computer Science

San Jose, CA

Sep. 2025 – April 2027

ICFAI Foundation For Higher Education

Bachelor of Technology in Data Science & Artificial Intelligence

Hyderabad, India

Aug. 2020 – Aug. 2024

EXPERIENCE

Machine Learning Engineer Intern

OSI Digital Pvt. Ltd.

Jan. 2024 – Jun. 2024

- Built production ML text classification services using transformer models (BERT, DistilBERT), improving accuracy by 25% and automating parsing of 5,000+ documents monthly with AWS Textract and Azure Document Intelligence.
- Deployed RAG chatbot architectures with vector databases (Pinecone, ChromaDB) and LangChain, reducing process hours by 75% and improving retrieval accuracy by 60%.
- Engineered FastAPI services integrating GPT-3.5/4 with custom prompt engineering, handling 1,000+ concurrent queries.
- Built MLOps pipelines with GitLab CI/CD, Docker, and automated testing (pytest), achieving 95% test coverage and reducing deployment time by 40%.

Software Engineering Intern

Tech Mahindra Ltd.

Jun. 2023 – Aug. 2023

- Developed backend microservices with LangChain and Python to process PDFs using OpenAI models, accelerating text extraction by 50%.
- Implemented monitoring with CloudWatch, improving system reliability by 35% and reducing MTTR by 60%.
- Deployed containerized services on AWS EC2 with auto-scaling, handling 100 to 5,000 requests per hour.

PROJECTS

CaseLawGPT – AI Legal Research Assistant | *Python, RAG, FAISS, LLMs, Docker, Streamlit, GCP*

2025

- Architected, deployed, and validated a Retrieval-Augmented Generation (RAG) system over 500+ U.S. Supreme Court opinions, reducing hallucination risk through citation-grounded responses generated from 8,000+ semantically indexed text chunks.
- Accelerated precedent retrieval by engineering a semantic search pipeline with sentence-transformer embeddings, FAISS, and SQLite, achieving sub-second top-K latency while supporting court- and date-filtered legal queries.
- Containerized and hosted the application on Google Cloud Platform using NVIDIA GPUs, integrating Ollama (Llama/Qwen models) and Streamlit, and validated end-to-end system performance through live endpoint testing by external users.

Ad-Click Predictor ML Pipeline | *Python, Scikit-learn, Pandas*

2024

- Built end-to-end ML pipeline with 89% accuracy using Logistic Regression and Random Forest with feature engineering.
- Encapsulated preprocessing and inference into reusable OOP modules with pickle serialization for CI/CD integration.